# PROJECT REPORT

## AI-Based Network Traffic Prediction

*Using Long Short-Term Memory (LSTM) Neural Networks*

By

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# 1. Abstract

The rapid expansion of internet-connected devices and the deployment of 5G technologies have led to an unprecedented increase in network traffic. Effectively predicting this traffic is crucial for proactive resource allocation, congestion control, and maintaining a high Quality of Service (QoS). This project proposes an AI-based approach for network traffic prediction using deep learning, specifically **Long Short-Term Memory (LSTM)** neural networks. By analyzing simulated time-series traffic data featuring daily and weekly seasonality alongside trend components, the LSTM model identifies complex underlying patterns. Experimental results confirm the effectiveness of this approach: the trained model achieved a **Mean Absolute Error (MAE) of 14.74 MB** and a **Root Mean Squared Error (RMSE) of 18.28 MB** on unseen test data, demonstrating the capability of deep learning to accurately forecast continuous traffic volumes while outperforming linear baselines and ensuring intelligent network management.

# 2. Introduction

In modern telecommunication networks, predicting traffic volume accurately is a fundamental requirement. Anticipating peaks and troughs allows internet service providers (ISPs) to reroute traffic smoothly, optimize bandwidth, scale cloud infrastructure dynamically, and minimize operational costs. Unlike traditional rule-based or purely statistical methods, modern Artificial Intelligence (AI) models can dynamically learn shifting temporal dependencies. This project demonstrates the usage of recurrent network architectures designed explicitly for sequence prediction to forecast data bursts and traffic trends reliably.

# 3. Problem Statement

Networks experience bandwidth under-utilization during off-peak hours and severe congestion during peak periods. Without knowing future traffic requirements, service providers manually scale resources, leading to either dropped packets and high latency (due to under-provisioning) or wasted energy and expenses (due to over-provisioning). The problem is to construct a scalable predictive model capable of forecasting future bandwidth consumption based on historical network usage logs.

# 4. Objectives

- To generate and analyze a realistic temporal network traffic dataset.
- To design a deep learning-based forecasting architecture utilizing sequence-to-sequence memory.
- To evaluate predictive performance against actual traffic outcomes using regression metrics.
- To showcase how artificial intelligence yields actionable intelligence for real-world telecom challenges.

# 5. Literature Review

## 5.1 ARIMA (Auto-Regressive Integrated Moving Average)

Traditional statistical approach for time-series forecasting. While interpretable and highly robust for linear trends, ARIMA struggles with high-dimensional noise and complex non-linear seasonality commonly found in modern bursty network traffic [1].

## 5.2 Support Vector Regression (SVR)

A machine learning model used to map non-linear data into higher dimensional spaces for regression. SVR is robust but lacks native memory, meaning it does not inherently learn sequential time-steps organically like recurrent models without heavy feature engineering [2].

## 5.3 LSTM Networks

The state-of-the-art for sequence data. Recurrent Neural Networks (RNNs) suffer from vanishing gradients over long sequences. LSTMs solve this via a gated cell structure (forget, input, and output gates), keeping long-term memory intact — perfect for recognizing weekly user traffic behavior alongside immediate daily spikes [3].

## 6. Methodology

This project follows a standard Data Science Pipeline:

1. **Data Generation:** Synthetic dataset mimicking real-world behaviors (combining global network growth trends, 24-hour cycles, and 168-hour weekly cycles).
2. **Preprocessing:** Data scaling using Min-Max Normalization to constrain feature space between [0, 1] ensuring rapid gradient descent convergence. Time-series sequences of a defined window width are prepared.
3. **Model Selection:** Building an LSTM neural network due to its empirical success in modeling dependent sequential tasks. Regularization techniques (Dropout) are implemented.
4. **Iterative Training:** Optimizing the Mean Squared Error (MSE) loss using the robust Adam optimizer.
5. **Testing & Inference:** Running predictions on unseen dataset chunks and performing inverse transformations to analyze error metrics.

## 7. System Architecture

1. **Input Layer:** Receives rolling windows of past network traffic (past 48 hours).
2. **Hidden LSTM Layers:** First LSTM layer (50 units, return_sequences=True) processes the sequence and outputs a full vector sequence to the next layer. Second LSTM layer (50 units, return_sequences=False) determines the most relevant temporal correlations.
3. **Dropout Layers:** 20% Dropout applied after each LSTM layer to prevent the system from memorizing training data, reducing overfitting by randomly freezing neurons during training passes.
4. **Dense Output Layer:** A fully connected layer (1 unit) that outputs a single continuous value representing the traffic in the next time step.

# 8. Algorithms Used

## 8.1 Long Short-Term Memory (LSTM)

Chosen over standard RNNs because network traffic has both short-term dependencies (traffic 1 hour ago) and long-term dependencies (traffic this exact time last week). The gated cell structure allows selective retention or forgetting of information, which is essential for handling both rapid daily spikes and slower weekly patterns.

## 8.2 Alternative Baselines

ARIMA and Linear Regression serve as conceptual baselines to contextualize the LSTM's performance. These methods lack native sequential memory and require extensive manual feature engineering to partially capture the non-linear, bursty behavior present in modern network traffic — a limitation that LSTM addresses inherently through its architecture.

# 9. Dataset Description

Since private ISP datasets are strictly confidential and restricted by privacy laws, this project utilizes a programmatically generated, highly realistic synthetic dataset. The dataset was designed with 2,500 hourly records — approximately 104 days — which provides roughly 14 complete weekly cycles, giving the LSTM model sufficient exposure to learn recurring weekly traffic patterns before encountering unseen test data.

- Features: Timestamp, Traffic Volume (MB)
- Observations: Hourly frequency capturing upward scaling trends, sinusoidal seasonal impacts, and Gaussian burst noise
- Scope: 2,500 hourly records (~104 days, ~14 weekly cycles)

# 10. Data Preprocessing Steps

1. **Missing Data Handling:** Simulated data is complete by construction. In a real-world deployment, linear interpolation would be applied to fill in any dropped log hours before further processing.

2. **Normalization:** The MinMaxScaler scales all values to [0, 1], reducing standard deviation discrepancies and preventing high-value outliers from dominating weight adjustments during backpropagation.
3. **Sequence Structuring:** Converting absolute values into relational input/target matrices where X = [$t_1$, $t_2$, ..., $t_{+8}$] and y = [$t_{+9}$]. A lookback window of 48 time steps (hours) was chosen to capture both immediate patterns and the prior two-day context.

## 11. Model Training and Testing

- Optimizer: Adam (Adaptive Moment Estimation), combining the best of AdaGrad and RMSProp.
- Loss Function: Mean Squared Error (MSE).

**Training Setup:** The dataset was split 80% for training and 20% for testing. Of the training portion, a further 10% was held back by Keras as a validation set (validation_split=0.1), which corresponds to approximately 8% of the total dataset. The model was trained for 15 epochs with a batch size of 32.

## 12. Performance Metrics

Since forecasting traffic is a regression problem, accuracy percentages are technically inappropriate. The following metrics validate the design:

- **Mean Absolute Error (MAE):** Represents the average absolute hourly deviation in Megabytes between the predicted and actual bandwidth. A lower MAE indicates consistently close predictions.
- **Root Mean Squared Error (RMSE):** Penalizes larger errors more heavily than MAE. A low RMSE indicates the model handles unpredictable traffic spikes safely.

**Actual Results:** MAE = 14.74 MB | RMSE = 18.28 MB

## 13. Results and Discussion

The LSTM model was evaluated on 500 hours of unseen test data. The model achieved a Mean Absolute Error of 14.74 MB and a Root Mean Squared Error of 18.28 MB, indicating that hourly traffic predictions deviated from actual values by approximately 14–18 MB on average — a reasonable margin given the synthetic noise component of up to ±15 MB introduced during data generation.

Visually, the predicted vs. actual plot confirms that the model successfully tracked the periodic crests and troughs of the daily traffic cycle and correctly followed the generalized upward trend across the test window. The LSTM predictions (shown as a dashed red line) closely matched the actual target traffic (orange) across both low-traffic troughs and high-traffic peaks from late March through early April 2025.

The model showed slightly larger deviations at sharp noise-driven spikes — instantaneous burst events that are inherently difficult to anticipate, as they are stochastic by design. This is reflected in the RMSE being somewhat higher than the MAE, which is expected behavior: RMSE penalizes these outlier deviations more heavily. Overall, the results demonstrate that the LSTM architecture reliably models complex seasonalities with less manual feature engineering than structural time-series approaches such as ARIMA.

Training converged rapidly — the MSE loss dropped from 0.029 to approximately 0.005 within just 3 epochs, and both training and validation loss curves remained stable and closely aligned throughout all 15 epochs, confirming that the Dropout regularization successfully prevented overfitting.

## 14. Limitations

The following limitations should be considered when interpreting the results of this project:

- Synthetic vs. Real Data: The dataset was programmatically generated and does not reflect the full complexity of real-world ISP traffic, which may include abrupt behavioral shifts, network outages, or geopolitical events that alter usage patterns unpredictably.

- Sensitivity to Noise: The model struggles with instantaneous burst events that fall outside the learned seasonal pattern, as evidenced by the higher RMSE relative to MAE. These are stochastic by nature and not fully predictable from historical windows alone.
- Anomaly Handling: Sudden large-scale traffic anomalies such as DDoS attacks were not explicitly modeled or labeled in the dataset. The model treats them as noise, meaning it is not natively suited for real-time anomaly detection without additional post-processing.
- Univariate Input: The model uses only historical traffic volume as input. Incorporating external signals (weather, events, server CPU load) could substantially reduce prediction error.

## 15. Future Scope

1. **Multivariate Forecasting:** Upgrading the dataset to include weather patterns, public event calendars, or CPU server metrics to aid prediction.
2. **Real-Time Streaming App:** Integrating with Apache Kafka to ingest live network switch data and output predictive actions to SDN (Software-Defined Network) controllers dynamically.
3. **Federated Learning:** Training models directly on decentralized edge routers to protect client privacy while enabling network-wide intelligence.

## 16. Conclusion

This project demonstrated that Deep Learning — specifically LSTM neural networks — offers an automated, scalable, and effective approach to network traffic prediction. The model successfully captured daily and weekly seasonalities in synthetic traffic data without requiring manual feature engineering. It achieved a Mean Absolute Error of 14.74 MB and an RMSE of 18.28 MB on unseen test data, with training converging in as few as 3 epochs and no evidence of overfitting. These results highlight the significant advantages of recurrent deep learning architectures over traditional statistical baselines such as ARIMA and SVR, particularly in handling the non-linear, bursty nature of real-world network traffic. The github link to view the project code is *https://github.com/safwanwaris/ai-network-traffic-prediction*

## 17. Why AI Outperforms Traditional Methods

1. **Non-Linear Dynamics:** Traditional statistical models expect traffic to broadly behave linearly. Real internet traffic is highly non-linear, dictated by spontaneous events (e.g., viral content, software update pushes). AI inherently maps non-linear topologies.
2. **Automatic Feature Extraction:** While statistical methods require human-in-the-loop stationary testing, differencing, and manual hyperparameter selection (P, D, Q in ARIMA), deep neural networks extract temporal features entirely implicitly.
3. **Scalability to Big Data:** Traditional approaches suffer significant computational slowdown on millions of rows. Deep Learning models, leveraging GPUs and batch processing, scale well on massive telecom datasets.

## 18. Real-World Applications

- **Smart Networks (SDN/NFV):** Utilizing predictions to dynamically instantiate virtual network machines where and when they are suddenly needed, maximizing network elasticity.
- **Cybersecurity & Anomaly Detection:** By establishing an AI-predicted normal baseline for upcoming traffic, sudden deviations — such as a DDoS attack — can instantly trigger security alerts before routers are overwhelmed.
- **Optimized Energy Consumption:** Sleep modes taking effect on cellular base stations or server racks during confidently predicted low-traffic periods, reducing a telecom provider's carbon footprint and operational costs.

## References

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